



Recovering copper from printed circuit boards (Photo courtesy: EMPA)

ment seem efficient and environmentally-friendly, but there are hidden dangers associated with them once these become e-waste. The harmful materials contained in electronics products, coupled with the fast rate at which we're replacing outdated units, pose a real danger to human health if electronics products are not properly processed prior to disposal.

Electronics products like computers and cellphones contain a lot of different toxins. For example, cathode ray tubes (CRTs) of computer monitors contain heavy metals such as lead, barium and cadmium, which can be very harmful to health if they enter the water system. These materials can cause damage

to the human nervous and respiratory systems. Flame-retardant plastics, used in electronics casings, release particles that can damage human endocrine functions. These are the types of things that can happen when unprocessed e-waste is put directly in landfill.

The scenario

The Basel Action Network (BAN) which works for prevention of globalisation of toxic chemicals has stated in a report that 50 to 80 per cent of e-waste collected by the US is exported to India, China, Pakistan, Taiwan and a number of African countries. This is done because cheaper labour is available for recycling in these countries. And in the

TABLE II
Discard Rate of Electronics Items

Item	Discard/replace rate
Mobile telephone	1 to 3 years
PC	Every 2 years
Camera	3 to 5 years
Television	10-15 years
Refrigerator	10-15 years
Washing Machine	10-15 years
IT accessories	Very fast

US, export of e-waste is legal.

e-waste recycling and disposal in China, India and Pakistan are highly polluting. Of late, China has banned import of e-waste. Export of e-waste by the US is seen as lack of responsibility on the part of Federal Government, electronics industry, consumers, recyclers and local governments towards viable and sustainable options for disposal of e-waste.

In India, recycling of e-waste is almost entirely left to the informal sector, which does not have adequate means to handle either the increasing quantities or certain processes, leading to intolerable risk for human health and the environment.

Dynamics of e-waste generation

Telecommunications and information technology are the fastest growing industries today not only in India but world over. Manufacturers' Association for Information Technology (MAIT) has collected the following statistics on the growth of electronics and IT equipment in India:

1. PC sales were over 7.3 million units during 2007-08, growing by 16 per cent. There is an installed base of over 25 million units.

2. The consumer electronics market is growing at the rate of 13-15 per cent annually. It has an installed base of 120 million TVs.

3. The cellular subscriber base was up by 96.86 per cent during 2007-08. Its installed base is estimated to cross 300 million mark by 2010.

With the unprecedented induction

TABLE I
e-Waste Toxins and Affected Body Parts

Components	Constituents	Affected body parts
Printed circuit boards	Lead and cadmium	Nervous system, kidney, liver
Motherboards	Berillium	Lungs, skin
Cathode ray tubes (CRTs)	Lead oxide, barium and cadmium	Heart, liver, muscles
Switches and flat-screen monitors	Mercury	Brain, skin
Computer batteries	Cadmium	Kidney, liver
Capacitors and transformers	Polychlorinated biphenyls (PCBs)	
Printed circuit boards, plastic	Brominated flame-retardant casings cable	
Cable insulation/coating	Polyvinyl chloride (PVC)	Immune system
Plastic housing	Bromine	Endocrine